

Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce
Department of Information Technology (B.Sc.I.T Semester V)
APA Practical

Practical – IX

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Practical 9: Hypothesis Testing and A/B Testing

Part A: Hypothesis Testing

Dataset: supermarket_sales.csv

1. Load the supermarket_sales.csv dataset.
2. Display the first five records.

```
import pandas as pd

import numpy as np

df = pd.read_csv(r'C:\Users\Hanoch\Desktop\College TY\Sem 5\Applied Business
  Analytics\practical 9\supermarket_sales - supermarket_sales.csv')

df.head()
```

	invoice_id	branch	city	customer_type	gender_customer	product_line	u
0	750-67-8428	A	Yangon	Member	Female	Health and beauty	
1	226-31-3081	C	Naypyitaw	Normal	Female	Electronic accessories	
2	631-41-3108	A	Yangon	Normal	Male	Home and lifestyle	
3	123-19-1176	A	Yangon	Member	Male	Health and beauty	
4	373-73-7910	A	Yangon	Normal	Male	Sports and travel	

3. Select the revenue column.
4. Check for missing values.

```
revenue_col = df['revenue']
revenue_col.isnull().sum()
```

✓ 0.0s

np.int64(0)

5. Calculate the mean revenue.

```
# mean revenue
mean_revenue = np.mean(revenue_col)
print("Mean Revenue is: ",mean_revenue)
```

✓ 0.0s

Mean Revenue is: 322.966749

6. Assume that the expected average revenue is ₹1,500.

7. Write the null hypothesis: [There is no significant difference between the true population mean and the expected value]

$H_0: \mu = 1500$

8. Write the alternative hypothesis: [There is significant difference between the true population mean and the expected value]

$H_1: \mu \neq 1500$

9. Set the significance level as 5%.

10. Perform a one-sample t-test.

```
# one sample t-test
from scipy.stats import ttest_1samp
t_statistic, p_value = ttest_1samp(revenue_col, mean_revenue)

# Print the results
print(f"T-statistic: {t_statistic:.4f}")
print(f"P-value: {p_value:.4f}")
```

✓ 0.0s

T-statistic: 0.0000
P-value: 1.0000

11. Calculate the p-value.

12. Compare the p-value with 0.05.

13. State whether the null hypothesis is rejected or not rejected.

14. Write the conclusion in simple business terms.

```
# one sample t-test
from scipy.stats import ttest_1samp
t_statistic, p_value = ttest_1samp(revenue_col, mean_revenue)

# Print the results
print(f"T-statistic: {t_statistic:.4f}")
print(f"P-value: {p_value:.4f}")

alpha = 0.05
if p_value <= alpha:
    print("Reject the null hypothesis.")
else:
    print("Fail to reject the null hypothesis.")
```

✓ 0.0s

T-statistic: 0.0000

P-value: 1.0000

Fail to reject the null hypothesis.

conclusion: [There is no significant difference between the expected mean and true mean means that the revenue which is calculated in business analytics is correct.]

Part B: A/B Testing

Dataset: ab_testing.csv

The dataset contains:

Group Users Conversions

A 1000 120

B 1000 150

- 1. Load the A/B testing dataset.
- 2. Display the data.

```
df = pd.read_csv(r'C:\Users\Hanoch\Desktop\ab_testing.csv')
df.head()
```

✓ 0.0s

	user_id	group	converted
0	user_1861	B	0
1	user_0354	A	0
2	user_1334	B	0
3	user_0906	A	0
4	user_1290	B	0

- 3. Calculate the conversion rate for Group A.
- 4. Calculate the conversion rate for Group B.

```
conversion_rates = df.groupby('group')['converted'].mean() * 100
print("Conversion Rate of A: ",conversion_rates['A'])
print("Conversion Rate of B: ",conversion_rates['B'])
```

✓ 0.0s

Conversion Rate of A: 12.0

Conversion Rate of B: 15.0

5. Compare the conversion rates of A and B.

[The Conversion Rate of A is 12.0% while the Conversion rate of B is 15.0%]

6. Write the null hypothesis: [there is no difference between the conversion rates of A and B. The new variant B did not prove anything]

$H_0: p_A = p_B$

7. Write the alternative hypothesis: [There is significant difference between the conversion rates of A and B]

$H_1: p_A \neq p_B$

8. Set the significance level as 5%.

9. Perform a two-proportion test.

10. Calculate the p-value.

11. Compare the p-value with 0.05.

12. State whether the null hypothesis is rejected or not rejected.

```
# proportion z-test
import statsmodels.api as sm
conversions = np.array([120,150])
total = np.array([1000,1000])

z_stat,p_val = sm.stats.proportions_ztest(
    count=conversions,nobs=total,alternative='two-sided'
)

print("Z-statistic: ",z_stat)
print("P-Value: ",p_value)

alpha = 0.05
if p_value<alpha:
    print("Reject the null hypothesis. The difference is statistically significant.")
else:
    print("Failed to reject the null hypothesis. The difference is not statistically significant.")
```

✓ 2.5s

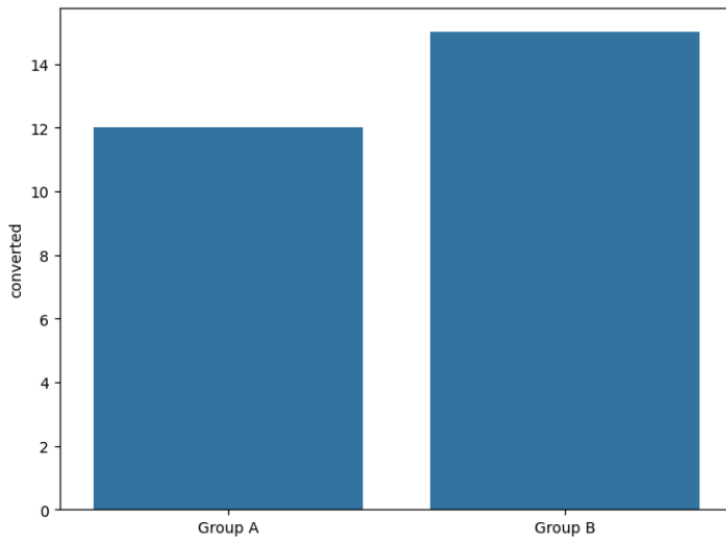
Z-statistic: -1.963049807622355
P-Value: 1.0
Failed to reject the null hypothesis. The difference is not statistically significant.

13. Create a bar chart to compare the conversion rates.

```
import seaborn as sns
import matplotlib.pyplot as plt

plt.figure(figsize=(8,6))
sns.barplot(x=['Group A','Group B'],y=conversion_rates)
plt.show()
```

✓ 0.1s



14. Write the conclusion based on the test result.

[Group A has lesser conversion rate as compared to Group B. Measures can be taken to improve the conversion rate of Group A]